

MLE for the Binomial, Exponential and Normal Distributions

Contents

1. Overview	2
2. The binomial distribution	3
2.1 From probability to likelihood	3
2.2 The derivation with numbers	3
2.3 The general formula	4
3. The exponential distribution	4
3.1 The distribution of waiting times	4
3.2 The likelihood of all the waits	5
3.3 The derivation	6
4. The normal distribution	6
4.1 One mouse, then two	6
4.2 The log-likelihood	7
4.3 The MLE of the mean	8
4.4 The MLE of the standard deviation	8
5. Dividing by n or by $n - 1$	9
5.1 Two formulas for one spread	9
5.2 Biased but consistent	9
6. Summary	11
7. Sources	11
8. Key terms	11

1. Overview

Key point: Following the maximum likelihood recipe by hand gives three famous answers: the binomial's $\hat{p}$ is the share of successes, the exponential's $\hat{\lambda}$ is one over the average waiting time, and the normal's $\hat{\mu}$ and $\hat{\sigma}$ are the mean and standard deviation of the data.

This Note follows the StatQuest lessons “Maximum Likelihood for the Binomial Distribution”, “Maximum Likelihood for the Exponential Distribution” and “Maximum Likelihood For the Normal Distribution” (Josh Starmer).

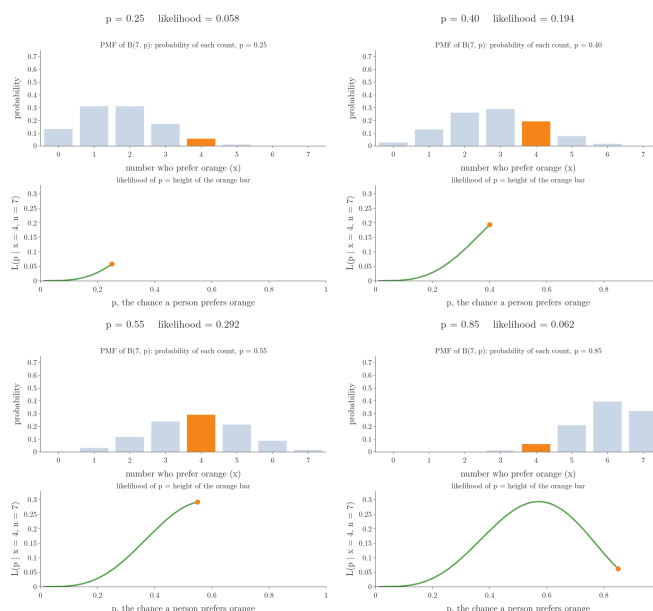


Figure 1: The PMF of $B(7, p)$ as p slides from 0.05 to 0.95. The orange bar is the probability of the observed count, 4 of 7; traced against p (bottom), it is the likelihood, highest at $p = 4/7$

The [maximum likelihood estimation Note](#) built the recipe: write the likelihood, take the log, differentiate, set the derivative to 0, solve. This Note runs it on three distributions, each in three steps (words, formula, small numbers):

- binomial: a share of yes answers (Section 2, Figure 1);
- exponential: waiting times (Section 3);
- normal: measurements such as weights (Section 4);
- why the normal's MLE variance divides by n , and how it differs from the sample variance with $n - 1$ (Section 5).

The Notebook checks every result by a grid search and against SciPy's built-in fitting.

2. The binomial distribution

Key point: If x of n trials are successes, the maximum likelihood estimate of the success probability is $\hat{p} = x/n$.

2.1 From probability to likelihood

Key point: The same binomial formula gives a probability when p is fixed and x varies, and a likelihood when x is fixed and p varies.

We ask 7 people whether they prefer orange or grape Fanta, and 4 say orange. Each answer is a Bernoulli trial, and the count of orange answers follows a binomial distribution $B(n, p)$ (see the [Bernoulli and binomial Note](#)):

$$P(x | n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$$

With $p = 0.5$ fixed, this is the probability that 4 of 7 people prefer orange: $35 \times 0.5^4 \times 0.5^3 = 0.273$. Now fix the data, $n = 7$ and $x = 4$, and let p vary. The right-hand side stays the same formula; only the reading changes (the [probability vs likelihood Note](#)). We write it as $L(p | n = 7, x = 4)$.

p	0.25	0.5	0.57	0.75
$L(p n = 7, x = 4)$	0.058	0.273	0.294	0.173

Figure 1 sweeps p from left to right. For each p the top panel shows the whole PMF; the bar at $x = 4$ is the likelihood. Traced against p , the bar heights form the curve at the bottom, which peaks a little above 0.57.

2.2 The derivation with numbers

Key point: Log, differentiate, set to 0: $4/p - 3/(1 - p) = 0$ gives $p = 4/7$.

1. **In words:** take the log of the likelihood, differentiate with respect to p , set the result to 0 and solve for p .
2. **Formula:** the log turns the product into a sum and brings the exponents down:

$$\ell(p) = \log \binom{7}{4} + 4 \log p + 3 \log(1 - p)$$

The first term has no p in it, so its derivative is 0. The derivative of $\log p$ is $1/p$. For $\log(1 - p)$ the chain rule (see the [derivatives Note](#)) gives $1/(1 - p) \times (-1)$:

$$\frac{d\ell}{dp} = \frac{4}{p} - \frac{3}{1 - p}$$

3. **Example:** set it to 0 and multiply both sides by $p(1 - p)$:

$$4(1 - p) - 3p = 0 \implies 4 - 7p = 0 \implies \hat{p} = \frac{4}{7} = 0.571$$

The likelihood there is 0.294, higher than at every p in the table.

2.3 The general formula

Key point: The same steps with letters give $\hat{p} = x/n$: successes divided by trials.

1. **In words:** for x successes in n trials, the MLE of p is the observed share of successes.
2. **Formula:** the log-likelihood and its derivative are

$$\ell(p) = \log \binom{n}{x} + x \log p + (n - x) \log(1 - p), \quad \frac{d\ell}{dp} = \frac{x}{p} - \frac{n - x}{1 - p}$$

Setting the derivative to 0 and multiplying by $p(1 - p)$: $x(1 - p) - (n - x)p = 0$. The terms $-xp$ and $+xp$ cancel, leaving $x - np = 0$:

$$\hat{p} = \frac{x}{n}$$

3. **Example:** 4 orange answers out of 7 give $4/7 = 0.571$; 40 out of 70 give the same 0.571.

The second derivative, $-x/p^2 - (n - x)/(1 - p)^2$, is negative for every p between 0 and 1, so this is a maximum.

The answer looks obvious once known: the best guess of a probability is the observed proportion. The derivation turns the intuition into a proof.

Extra: With 40 of 70 the peak sits at the same 0.571, but the likelihood curve is much narrower: more data pins p down more tightly. The Notebook draws both curves.

3. The exponential distribution

Key point: For waiting times $x_1, \dots, x_n$, the MLE of the rate is $\hat{\lambda} = n / \sum x_i = 1/\bar{x}$: one over the average wait.

3.1 The distribution of waiting times

Key point: $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. The rate λ is how many events happen per unit time on average; the average wait is $1/\lambda$.

The **exponential distribution** models the time between random events: the wait for the next text message, the time until the next person opens a web page. The distribution is right-skewed: short waits are common, long waits rare. The [sampling distribution and CLT Note](#) used it as a skewed population.

1. **In words:** the density starts at height λ at $x = 0$ and falls off exponentially. The **rate parameter** λ is the average number of events per unit of time.
2. **Formula:**

$$f(x | \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

3. **Example:** with $\lambda = 2$ events per second, on average one event every $1/2$ second; the density at $x = 1$ second is $2e^{-2} = 0.27$. With $\lambda = 0.5$, one event every 2 seconds, and the density at 1 second is $0.5e^{-0.5} = 0.30$.

Figure 2 (left) shows the three curves for $\lambda = 0.5, 1$ and 2 . A large rate means a tall start and a fast drop: short waits.

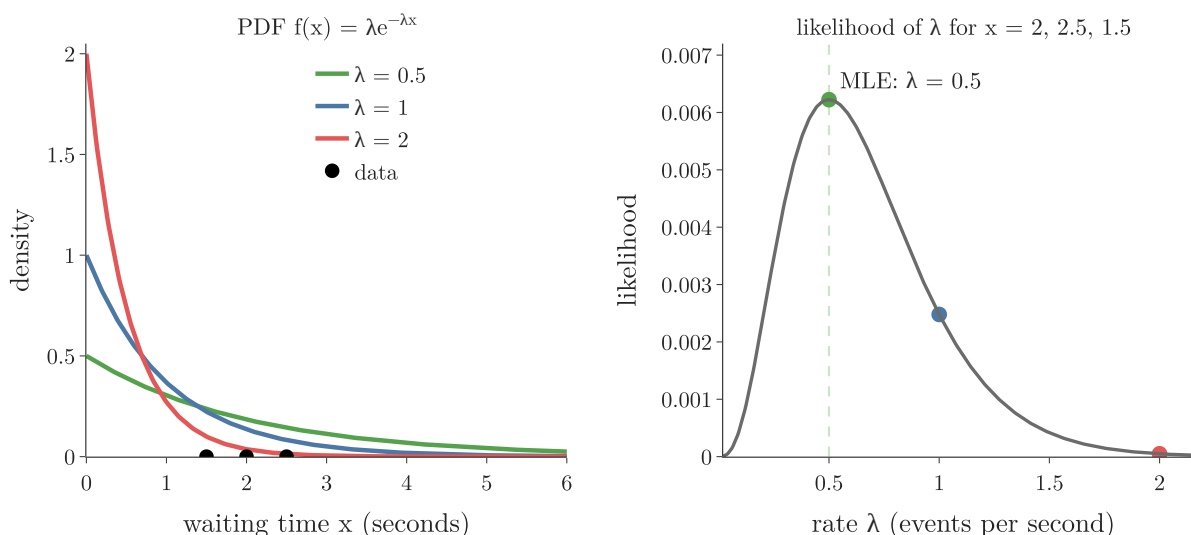


Figure 2: Left: exponential PDFs for three rates and the three waiting times. Right: the likelihood of the rate for those waiting times, highest at 0.5

3.2 The likelihood of all the waits

Key point: Multiplying the n densities collects the λ 's into λ^n and the exponents into $-\lambda \sum x_i$.

1. **In words:** each waiting time contributes the height of the curve above it. For independent waits the heights multiply, as in the [maximum likelihood estimation Note](#).
2. **Formula:** pull the n factors λ out of the product and add the exponents ($e^a e^b = e^{a+b}$):

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda(x_1 + \dots + x_n)}$$

3. **Example:** three waits between views of a web page: $x_1 = 2, x_2 = 2.5, x_3 = 1.5$ seconds, with sum 6. For $\lambda = 1$, $L = 1^3 e^{-6} = 0.0025$; for $\lambda = 0.5$, $L = 0.5^3 e^{-3} = 0.0062$.

3.3 The derivation

Key point: $\ell(\lambda) = n \log \lambda - \lambda \sum x_i$; its derivative $n/\lambda - \sum x_i$ is 0 at $\hat{\lambda} = n / \sum x_i$.

1. **In words:** the log turns λ^n into $n \log \lambda$ and the exponential into its exponent. Then differentiate and set to 0.

2. **Formula:**

$$\ell(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i, \quad \frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0 \implies \hat{\lambda} = \frac{n}{\sum_i x_i} = \frac{1}{\bar{x}}$$

3. **Example:** $\hat{\lambda} = 3/(2 + 2.5 + 1.5) = 3/6 = 0.5$ events per second. The average wait is 2 seconds, so the rate is one event every 2 seconds.

The second derivative is $-n/\lambda^2 < 0$, so this is the peak of Figure 2 (right). The fitted model is the green curve on the left, $f(x) = 0.5e^{-0.5x}$.

Python: SciPy's `fit` method uses maximum likelihood by default (its documentation says so).

```
import numpy as np
from scipy import stats
```

```
waits = np.array([2, 2.5, 1.5])
# floc=0 fixes the start of the curve at 0
loc, scale = stats.expon.fit(waits, floc=0)
scale          # 2.0: SciPy uses scale = 1 / lambda
1 / scale      # 0.5, the same as 3 / 6
```

SciPy describes the exponential by its **scale**, the average wait $1/\lambda$, rather than by the rate (`scipy.stats.expon` documentation). SciPy also has a shift parameter `loc`; without `floc=0` SciPy fits the shift too, and the Notebook shows the curve's start then moves to the smallest wait, 1.5.

4. The normal distribution

Key point: The MLE of μ is the mean of the data, and the MLE of σ is the standard deviation computed with n in the denominator.

4.1 One mouse, then two

Key point: With one observation (one recorded value) the likelihood is the height of the curve above it; with two, the product of two heights.

The normal PDF (see the [normal distribution Note](#)) has two parameters, the mean μ (location) and the standard deviation σ (width):

$$f(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

We weigh one mouse: 32 grams. With $\sigma = 2$ fixed, the likelihood of $\mu = 28$ is $f(32 | 28, 2) = 0.027$; of $\mu = 30$ it is 0.121; of $\mu = 32$ it is 0.199, the highest. With one point, the best curve is centred on it.

With two mice, 32 and 34 grams, the likelihood of a curve is the product of its two heights, because the weighings are independent. For $\mu = 28$, $\sigma = 2$: $0.027 \times 0.0022 = 6.0 \times 10^{-5}$. With n mice it is a product of n heights.

Extra: One point is not enough to estimate σ . With μ on the single point, the likelihood is the height of the curve at its own centre, $f(x | x, \sigma) = 1/(\sigma\sqrt{2\pi})$, which grows without limit as $\sigma \rightarrow 0$: there is no best σ . The Notebook prints 0.20, 0.80, 3.99 and 39.89 for $\sigma = 2, 0.5, 0.1$ and 0.01. The same runaway returns in the [Gaussian mixture models Note](#).

4.2 The log-likelihood

Key point: The log of one normal density is $-\frac{1}{2}\log(2\pi) - \log\sigma - (x - \mu)^2/(2\sigma^2)$; summing over n points gives the log-likelihood.

1. **In words:** the log splits each density into three pieces: a constant, minus $\log\sigma$, and minus the squared distance from the mean divided by $2\sigma^2$.
2. **Formula:** using $\log(1/a) = -\log a$, $\log\sqrt{a} = \frac{1}{2}\log a$ and $\log e^b = b$,

$$\log f(x_i | \mu, \sigma) = -\frac{1}{2}\log(2\pi) - \log\sigma - \frac{(x_i - \mu)^2}{2\sigma^2}$$

Adding the n terms, the first two pieces appear n times:

$$\ell(\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^n(x_i - \mu)^2$$

3. **Example:** for the five mice of the [maximum likelihood estimation Note](#), 29, 31, 32, 33, 35, at $\mu = 32$, $\sigma = 2$: the squared distances are 9, 1, 0, 1, 9, with sum 20, so

$$\ell = -2.5 \times 1.838 - 5 \times 0.693 - \frac{20}{8} = -4.595 - 3.466 - 2.5 = -10.56$$

Figure 3 shows ℓ over both parameters as a contour map. The map has a single peak. The dashed lines are the two one-parameter searches of the [maximum likelihood estimation Note](#); both cross at the peak.

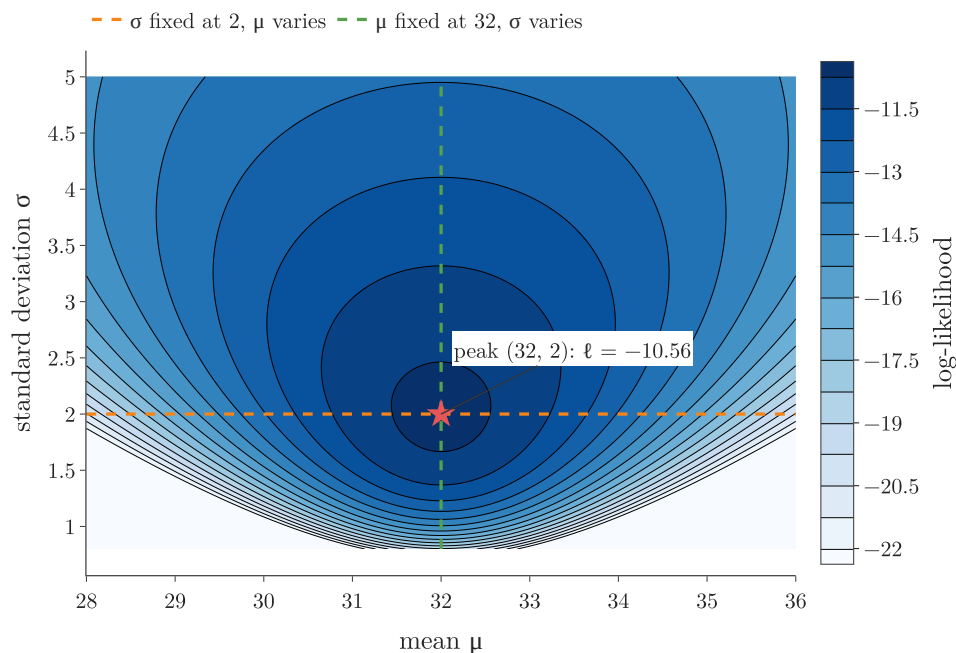


Figure 3: Contour map of the normal log-likelihood of the five mouse weights over the mean and the standard deviation, with a single peak at (32, 2)

4.3 The MLE of the mean

Key point: Differentiating with respect to μ (with σ held constant) and setting to 0 gives $\hat{\mu} = \bar{x}$.

1. **In words:** only the last term of ℓ contains μ . Treat σ as a constant, take the partial derivative with respect to μ (see the [partial derivatives and gradients Note](#)), set it to 0.
2. **Formula:** by the chain rule, the derivative of $(x_i - \mu)^2$ with respect to μ is $2(x_i - \mu) \times (-1)$, so

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu) = \frac{1}{\sigma^2} \left(\sum_i x_i - n\mu \right) = 0 \implies \hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}$$

3. **Example:** $\hat{\mu} = (29 + 31 + 32 + 33 + 35)/5 = 160/5 = 32$ grams.

The answer does not depend on σ : whatever the width, the best centre is the average.

4.4 The MLE of the standard deviation

Key point: Differentiating with respect to σ (with μ held at $\hat{\mu}$) gives $\hat{\sigma}^2 = \sum (x_i - \bar{x})^2 / n$.

1. **In words:** now treat μ as a constant. Two terms of ℓ contain σ : $-n \log \sigma$ and the sum divided by $2\sigma^2$.

2. **Formula:** the derivative of $-n \log \sigma$ is $-n/\sigma$. Writing $1/(2\sigma^2)$ as $\frac{1}{2}\sigma^{-2}$, its derivative is $-\sigma^{-3}$, so the minus sign in front makes the second term positive:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3} \sum_{i=1}^n (x_i - \mu)^2 = 0$$

Multiply by σ^3 , move $n\sigma^2$ to the other side, divide by n , and put in $\mu = \hat{\mu}$:

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad \hat{\sigma} = \sqrt{\hat{\sigma}^2}$$

3. **Example:** the squared distances from 32 add up to 20, so $\hat{\sigma}^2 = 20/5 = 4$ and $\hat{\sigma} = 2$ grams.

So the maximum likelihood normal curve for the mice is $N(32, 2^2)$, exactly where the searches in the [maximum likelihood estimation Note](#) peaked.

```
Python: stats.norm.fit returns the two maximum likelihood estimates.
mice = np.array([29, 31, 32, 33, 35])
stats.norm.fit(mice)      # (32.0, 2.0)
mice.mean(), mice.std()  # (32.0, 2.0): NumPy's std divides by n
mice.std(ddof=1)         # 2.236: divides by n - 1
```

5. Dividing by n or by $n - 1$

Key point: The MLE variance divides by n and is slightly too small on average; the sample variance divides by $n - 1$ and is right on average. The gap vanishes as n grows.

5.1 Two formulas for one spread

Key point: For the five mice, the MLE variance is 4 and the sample variance is 5.

The [measures of dispersion Note](#) (Section 4.3) defined the sample variance s^2 with $n - 1$ in the denominator, Bessel's correction. Maximum likelihood gives n instead.

1. **In words:** both formulas add the squared distances from the sample mean; they divide by different counts.
2. **Formula:**

$$\hat{\sigma}_{\text{ML}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{n}{n-1} \hat{\sigma}_{\text{ML}}^2$$

3. **Example:** the sum is 20. $\hat{\sigma}_{\text{ML}}^2 = 20/5 = 4$; $s^2 = 20/4 = 5$.

5.2 Biased but consistent

Key point: On average the MLE variance equals $(n-1)/n \times \sigma^2$, so it is biased low; since $(n-1)/n \rightarrow 1$, it is still consistent.

An estimator is **unbiased** if its average over many samples equals the true value, and **biased** if it is systematically off. The intuition: the sample mean $\bar{x}$ sits in the middle of its own sample, so the observations are always a little closer to $\bar{x}$ than to the true mean μ , and squared distances measured from $\bar{x}$ come out a little too small. The identity $\sum_i (x_i - \bar{x})^2 = \sum_i (x_i - \mu)^2 - n(\bar{x} - \mu)^2$ makes the gap exact. The measures of dispersion Note showed by simulation that dividing by n comes out too small on average. The size of the gap is a simple factor:

1. **In words:** averaged over many samples of size n , the MLE variance is the true variance times $(n-1)/n$.

2. **Formula:**

$$\text{average of } \hat{\sigma}_{\text{ML}}^2 = \frac{n-1}{n} \sigma^2, \quad \text{average of } s^2 = \sigma^2$$

3. **Example:** for samples of 5 mice from a population with $\sigma^2 = 4$, the MLE variance averages $4/5 \times 4 = 3.2$, while s^2 averages 4. With $n = 100$ the factor is 0.99, and the difference hardly matters. The Notebook draws 100,000 samples of 5 and gets 3.21 and 4.01.

Extra: Why the factor is $(n-1)/n$. Write $x_i - \mu = (x_i - \bar{x}) + (\bar{x} - \mu)$, square, and add over i . The cross term $2(\bar{x} - \mu) \sum_i (x_i - \bar{x})$ is 0, because the distances from the mean add up to 0. So

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n (x_i - \mu)^2 - n(\bar{x} - \mu)^2$$

Averaged over many samples, each $(x_i - \mu)^2$ is σ^2 , and $(\bar{x} - \mu)^2$ is the variance of the sample mean, σ^2/n (the [sampling distribution and CLT Note](#)). The right-hand side averages $n\sigma^2 - \sigma^2 = (n-1)\sigma^2$; dividing by n gives the factor.

So maximum likelihood does not promise an unbiased estimate. Maximum likelihood promises the parameters that make the observed data most likely, and, as n grows, an estimate that converges to the truth (consistency, from the [maximum likelihood estimation Note](#)).

6. Summary

Distribution	Parameter	Log-likelihood ℓ	MLE	Example
Binomial $B(n, p)$	p	$x \log p + (n - x) \log(1 - p) + \text{const}$	$\hat{p} = x/n$	4 of 7: 0.571
Exponential	λ	$n \log \lambda - \lambda \sum x_i$	$\hat{\lambda} = 1/\bar{x}$	waits 2, 2.5, 1.5: 0.5
Normal	μ	$-n \log \sigma - \sum (x_i - \mu)^2 / (2\sigma^2) + \text{const}$	$\hat{\mu} = \bar{x}$	mice: 32
Normal	σ	same	$\hat{\sigma}^2 = \sum (x_i - \bar{x})^2 / n$	mice: $\hat{\sigma} = 2$

- Every derivation follows the same recipe: product of densities, log, derivative, set to 0, solve.
- The binomial MLE is the observed share of successes.
- The exponential distribution $\lambda e^{-\lambda x}$ models waiting times; its MLE rate is one over the average wait.
- The normal MLEs are the sample mean and the standard deviation with n in the denominator.
- The MLE variance is biased low by the factor $(n - 1)/n$; the sample variance with $n - 1$ is unbiased. Both agree for large n .
- SciPy's `fit` methods compute maximum likelihood estimates by default.

7. Sources

- Starmer, J. “Maximum Likelihood for the Binomial Distribution”, StatQuest (statquest.org). Section 2.
- Starmer, J. “Maximum Likelihood for the Exponential Distribution”, StatQuest (statquest.org). Section 3.
- Starmer, J. “Maximum Likelihood For the Normal Distribution”, StatQuest (statquest.org). Section 4.
- SciPy documentation: `scipy.stats.expon` (scale = $1/\lambda$) and `rv_continuous.fit` (maximum likelihood by default). Section 3.3.

8. Key terms

Term	Meaning
Exponential distribution	The distribution of waiting times between random events, $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$

Term	Meaning
Rate parameter λ	The average number of events per unit time; the average wait is $1/\lambda$
Scale (in SciPy)	SciPy's parameter for the exponential distribution, the average wait $1/\lambda$
MLE of a binomial p	$\hat{p} = x/n$, the observed share of successes
MLE of an exponential rate	$\hat{\lambda} = n / \sum x_i = 1/\bar{x}$
MLE of a normal distribution	$\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \sum (x_i - \bar{x})^2 / n$
Unbiased estimator	An estimator whose average over many samples equals the true value
Biased estimator	An estimator that is systematically too high or too low on average